



Storm Shelter Challenge



How do we build a shelter that stays standing and dry in a storm?

Big storms bring strong wind and heavy rain. People need shelters that stay standing and keep them dry. Can you build one?

TIME

60 minutes

GROUPS

Groups of 3–4

AGES

Starter 6–8 · Explorer 9–11

Learning goals

- ✓ I can build a shelter that stands up by itself.
- ✓ I can test my shelter with wind and rain, fairly.
- ✓ I can improve my design after testing it.

STEAM in this activity

S

SCIENCE

Wind pushes on buildings; rain runs off slopes.

T

TECHNOLOGY

A fan and spray bottle model a storm safely.

E

ENGINEERING

Design, test, improve: the engineering cycle.

A

ART

Design and colour a storm-proof dream house.

M

MATHS

Count bricks and put fan speeds in order.



Before the lesson

- ☐ Put the same number of LEGO bricks in a bag for each group.
- ☐ Set up the wind station and the rain station apart, so water stays away from the fan.
- ☐ Cut card for roofs and small tissue squares.
- ☐ Check the fan has three speeds and a safe guard.

⚠ Safety

- Keep small bricks away from little children.
- Keep fingers away from the fan blades.
- An adult switches the fan on and off.
- Keep water far away from the fan and plugs.
- Wipe up spills so nobody slips.



Kit per group

- | | | |
|---|---|---|
| ☐  LEGO bricks and a base plate | ☐  Card for a roof | ☐  A small piece of tissue |
| ☐  The same number of bricks for two shelters | ☐  A desk fan with three speeds | ☐  Card for a flat roof and a sloped roof |
| ☐  A spray bottle of water | ☐  Two dry pieces of tissue | |

Running order (60 minutes)

0–10 min	Story: what happens to buildings in storms?
10–25 min	Build: shelter with a roof and tissue inside.
25–40 min	Test 1: tall or wide against the fan.
40–55 min	Test 2: flat or sloped roof in the "rain".
55–60 min	Share: what made the strongest, driest shelter?

The investigations

Start here: build: Build your shelter

Children record: Does it stand up by itself?; Draw your shelter *Explorers also:* How many bricks did you use?

What scientists often see: Engineers design buildings to be strong in storms. They test models first, just like you.

Test 1: Tall or wide?: Which shelter stands up to stronger wind?

Children record: Which fan speed knocked it over?; Draw your shelter

What scientists often see: Tall, thin buildings catch more wind up high and tip more easily. Low, wide buildings are more stable. Engineers make tall buildings extra strong.

Test 2: Flat roof or sloped roof?: Which roof keeps the tissue drier?

Children record: How is the tissue inside?; Draw your shelter

What scientists often see: Rain runs off a sloped roof. On a flat roof it can sit in puddles and leak through. That is why many houses have sloped roofs.

Talk about it

- What made your shelter strong?
- What would you change if you built it again?
- Why do many houses in rainy places have sloped roofs?

What to look for

Starter: Builds a shelter that stands and says what knocked it over.

Explorer: Uses results to explain one change that would make the shelter stronger or drier.

Art corner

My storm-proof house. Draw and colour your dream storm-proof house. Label the strong walls and the roof.

Maths corner

Starter: Count the bricks in one wall. How many walls? Draw them.

Explorer: Bricks in one wall × number of walls = _____. Which design used fewer bricks?

Real-world link

Engineers test model buildings with strong wind before the real building is made.

Going further

Add a triangle brace to the tall shelter and test it again with the fan.

